

Dataset Assessment

Project: E-Commerce Sales & Customer Analytics Dashboard

1. Introduction

The Indian E-Commerce Sales Analytics Dataset was obtained from Kaggle and used as the primary data source for the E-Commerce Sales & Customer Analytics Dashboard developed in Power BI. The purpose of this assessment is to understand the structure, quality, completeness, and suitability of the datasets before data cleaning, transformation, modelling, and analysis.

2. Data Source

Item	Details
Dataset	Indian E-Commerce Sales Analytics Dataset
Source	Kaggle
Dataset Provider	Jatin Khandelwal
Main File	sales.csv
Other Files Used	products.csv, customers.csv
Tool Used	Power BI / Power Query
Purpose	E-commerce sales and customer analytics

3. Dataset Overview

Dataset	Records	Size	Purpose
Sales	250,000	5.81 MB	Transaction and order-level analysis
Products	2,000	172.35 kB	Product, category, brand, pricing and rating analysis
Customers	40,000	39.66 MB	Customer, demographic and segmentation analysis

4. Dataset Contents

Sales: Order ID, Customer ID, Product ID, Order Date, Order Time, Delivery Date, Quantity, Unit Price, Order Value, Shipping Cost, Coupon Discount, Total Amount, Payment Mode, Order Status, Rating, Review Text, City, State, Customer Age and Customer Age Group.

Products: Product ID, Product Name, Category, Brand, Original Price, Discount Percentage, Discount Amount, Selling Price, Stock Quantity, Weight, Average Rating and Total Reviews.

Customers: Customer ID, Customer Name, Gender, Age, Age Group, Date of Birth, City, State, Pincode, Registration Date, Customer Tier, Total Orders and Total Spent.

5. Data Quality Assessment

Assessment Area	Observation
Missing Values	Blank and null values were checked in important columns.
Duplicate Records	Duplicate-looking records were reviewed, particularly Order_ID.
Data Types	Date, numeric, text and categorical columns were checked for appropriate data types.
Categorical Values	Category, payment mode, order status, gender and customer tier values were reviewed for consistency.
Date Fields	Order Date and Delivery Date were checked for valid date values.
Key Fields	Order_ID, Customer_ID and Product_ID were checked for relationship suitability.
Numerical Fields	Quantity, prices, discounts, shipping costs and total amounts were checked for valid numerical values.
Relationships	Common keys between Sales, Products and Customers were reviewed.

6. Important Observation: Duplicate Order IDs

A repeated Order_ID does not necessarily indicate a duplicate transaction. An individual order can contain multiple products, so the same Order_ID may appear in multiple rows with different Product_ID values. Therefore, Order_ID should not be removed simply because it appears multiple times. For order-level KPIs such as Total Orders, DISTINCTCOUNT (Order_ID) should be used.

Example: ORD1001 may appear on multiple rows when the order contains several products.

7. Data Cleaning Requirements

- Handle missing and null values.
- Check genuine duplicate records.
- Correct inappropriate data types.
- Standardize categorical values.
- Validate date columns.
- Check numerical values for inconsistencies.
- Ensure key columns are suitable for relationships.
- Create a dedicated Date table for time-based analysis.
- Prepare datasets for Power BI data modelling.

8. Data Suitability

The datasets contain sufficient information for the planned e-commerce analytics project. They support analysis of sales, products and categories, customers and segments, geography, order status, payment methods, cancellations, returns, delivery performance, discounts and pricing.

9. Assessment Conclusion

The Indian E-Commerce Sales Analytics Dataset was assessed before data cleaning and Power BI modelling. The assessment identified areas requiring attention, including missing values, duplicate-looking records, data types, categorical consistency, and relationship keys. After the required cleaning and transformation, the datasets were considered suitable for developing the E-Commerce Sales & Customer Analytics Dashboard in Power BI.

Note: Data assessment describes the condition of the raw data before transformation, while the Data Cleaning section documents the actual changes made to resolve the identified issues.